

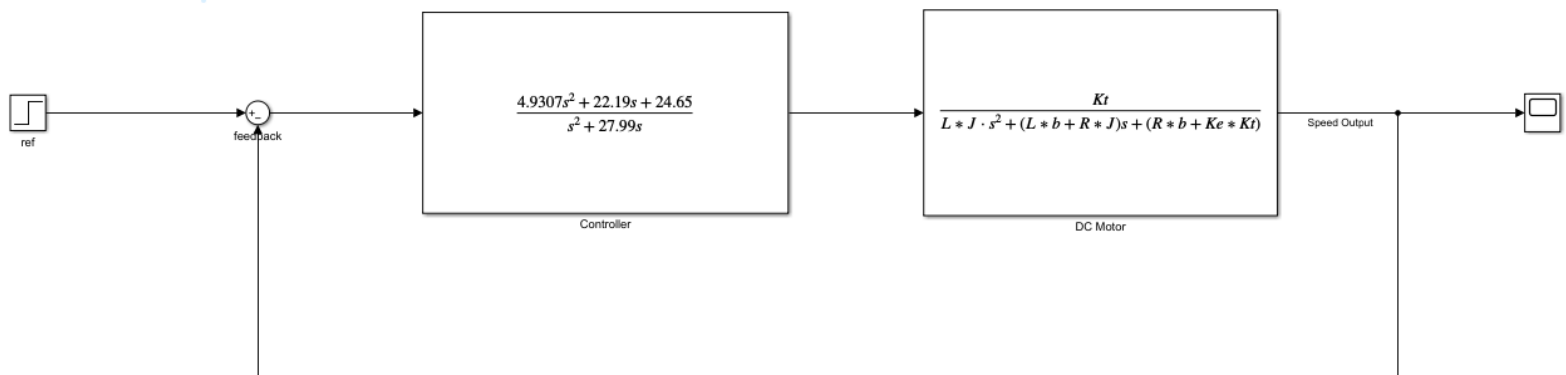
Classical PI–Lead Controller Design for DC Motor Speed Control Using MATLAB/Simulink

A Simulation-Based Control Systems Project

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Software Used

- MATLAB R2025b
- Simulink

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Abstract

This project presents the modeling, design, and performance evaluation of a classical PI + Lead controller for the speed control of a separately excited DC motor using MATLAB and Simulink. A mathematical model of the motor was first derived from its electrical and mechanical governing equations and represented as a transfer function relating the applied armature voltage to the motor angular speed. The open-loop response was analyzed to assess the motor's inherent dynamic characteristics and to establish realistic controller design requirements. An initial PI controller was investigated but was found insufficient to simultaneously satisfy the desired transient and steady-state specifications. Consequently, a PI + Lead controller was designed using the root locus method to improve the transient response while preserving zero steady-state error. The resulting controller achieved stable closed-loop operation with a settling time of approximately 2 s, an overshoot below 5%, and zero steady-state error. The controller was further validated through simulations involving time-varying reference tracking, disturbance rejection under external load torques, and robustness analysis against variations in key motor parameters. The results demonstrated satisfactory performance for moderate disturbances and parameter uncertainties while highlighting the influence of actuator saturation and increased rotor inertia on the closed-loop response. Overall, the project illustrates a complete classical control design workflow and demonstrates the effectiveness of the proposed controller for light-duty DC motor speed regulation.

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1. Introduction

1. Background

Direct current (DC) motors are among the most widely used actuators in modern electromechanical systems because of their simple construction, ease of control, and reliable speed regulation capabilities. They are commonly employed in applications such as robotic manipulators, conveyor systems, electric vehicles, automated manufacturing equipment, and numerous small-scale mechanical devices. In many of these applications, maintaining a desired rotational speed despite changes in operating conditions is essential for achieving accurate, stable, and efficient system performance.

Operating a DC motor in an open-loop configuration by applying a fixed armature voltage is generally insufficient for precise speed regulation. Variations in mechanical load, friction, and motor parameters can significantly affect the motor speed, causing deviations from the desired operating point. Consequently, feedback control techniques are employed to continuously monitor the motor speed and adjust the applied voltage, thereby reducing the tracking error and improving the overall system performance.

2. Problem Statement

This project addresses the problem of regulating the angular speed of a small light-duty DC motor using a classical feedback controller. The controller receives the error between the desired and measured motor speeds and adjusts the armature voltage supplied to the motor in order to minimize this error. Besides accurately tracking the reference speed, the controller should maintain satisfactory transient performance, reject external load disturbances, and preserve acceptable operation in the presence of moderate variations in motor parameters.

3. Project Objective

The objective of this project is to develop a mathematical model of a DC motor and design a classical feedback controller capable of regulating its angular speed according to predefined performance specifications. The work begins by deriving the motor transfer function from its electrical and mechanical governing equations, followed by an analysis of the open-loop system to understand its dynamic characteristics and to select suitable motor parameters.

A feedback controller is then designed using classical control techniques and evaluated through time-domain simulations in MATLAB. The controller is further validated by examining its transient response, disturbance rejection capability, and robustness against parameter variations using Simulink. The overall goal is to obtain a controller that provides stable, accurate, and physically realistic speed regulation while demonstrating a systematic engineering design process.

4. Design Requirements

The controller was designed to satisfy a set of performance specifications representative of a small light-duty DC motor speed control application. The primary objectives were to achieve a settling time not exceeding approximately 2 seconds, limit the maximum overshoot to less than 5%, eliminate the steady-state error for a step speed reference, and ensure stable closed-loop operation. In addition, the controller was expected to produce a physically realistic control effort that remains compatible with the assumed motor supply voltage and practical operating conditions.

5. Software Environment

MATLAB and Simulink were used throughout the development and evaluation of the control system. MATLAB was employed for mathematical modeling, transfer function analysis, controller design, pole analysis, and transient response evaluation. Simulink was used to construct the complete closed-loop control system, simulate external load disturbances, investigate actuator saturation effects, and assess the controller performance under different operating conditions.

6. Project Scope

This work focuses on the simulation-based design and evaluation of a classical controller for DC motor speed regulation. The study includes mathematical modeling, open-loop analysis, controller design, disturbance rejection analysis, and robustness evaluation under parameter uncertainties. The motor is represented by a linear mathematical model derived from its governing equations, while practical considerations such as actuator voltage saturation are incorporated during the validation stage.

The project does not include hardware implementation, experimental validation, sensor noise, PWM motor driver dynamics, or nonlinear effects such as Coulomb friction and magnetic saturation. These aspects are considered beyond the scope of the present work and represent possible extensions for future development.

2. DC Motor Modeling

1. Physical Model

The DC motor considered in this project consists of an electrical armature circuit coupled to a mechanical rotating shaft. The armature voltage generates an electrical current that produces an electromagnetic torque, causing the rotor to rotate. During operation, the motor experiences viscous friction and generates a back electromotive force (back EMF), both of which influence its dynamic behavior.

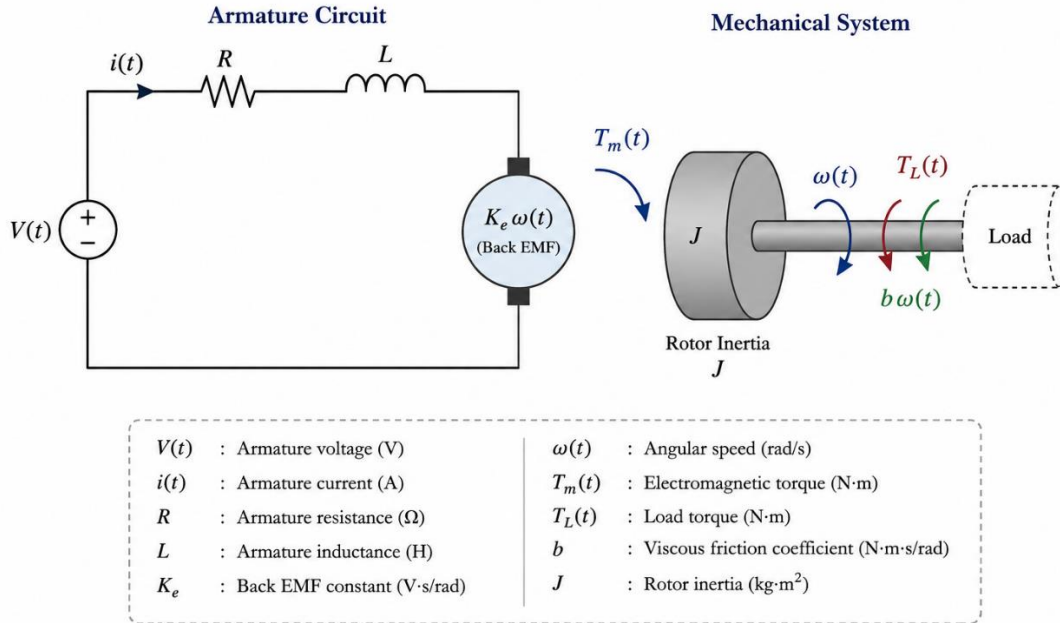


Figure 2.1: Schematic representation of a DC motor showing the electrical armature circuit and the mechanical rotational system.

To facilitate controller design, the motor is represented by a linear mathematical model relating the applied armature voltage to the resulting angular speed.

2. Modeling Assumptions

To simplify the mathematical model while preserving the dominant dynamics of the motor, the following assumptions were made:

- The motor operates in its linear region.
- Magnetic saturation is neglected.
- Armature reaction is neglected.
- The field flux is constant.
- Viscous friction is proportional to angular velocity.
- Coulomb friction and backlash are neglected.
- Motor parameters remain constant during operation.
- The load torque is assumed to be zero during the derivation of the nominal transfer function.

These assumptions are commonly adopted in classical DC motor modeling and provide sufficient accuracy for controller design and simulation.

3. Governing Equations

The electrical dynamics of the armature circuit are described by:

$$V(t) = L \frac{di(t)}{dt} + Ri(t) + K_e \omega(t)$$

Where $V(t)$ is the armature voltage, $i(t)$ is the armature current, R is the armature resistance, L is the armature inductance, K_e is the back EMF constant, $\omega(t)$ is the angular velocity.

The mechanical dynamics are governed by Newton's second law for rotational motion:

$$J \frac{d\omega(t)}{dt} + b\omega(t) = K_t i(t)$$

Where J is the rotor inertia, b is the viscous friction coefficient, K_t is the motor torque constant.

4. Transfer Function Derivation

From the mechanical equation, the armature current can be expressed as:

$$i(t) = \frac{J\dot{\omega}(t) + b\omega(t)}{K_t}$$

Differentiating with respect to time gives:

$$\frac{di(t)}{dt} = \frac{J\ddot{\omega}(t) + b\dot{\omega}(t)}{K_t}$$

Substituting these expressions into the electrical equation results in:

$$V(t) = \frac{L}{K_t} (J\ddot{\omega} + b\dot{\omega}) + \frac{R}{K_t} (J\dot{\omega} + b\omega) + K_e \omega$$

Collecting like terms yields:

$$LJ\ddot{\omega} + (RJ + Lb)\dot{\omega} + (Rb + K_t K_e)\omega = K_t V$$

Applying the Laplace transform under zero initial conditions gives:

$$[LJs^2 + (RJ + Lb)s + (Rb + K_t K_e)]\Omega(s) = K_t V(s)$$

Therefore, the voltage-to-speed transfer function becomes:

$$G_v(s) = \frac{\Omega(s)}{V(s)} = \frac{K_t}{LJs^2 + (RJ + Lb)s + (Rb + K_t K_e)}$$

This transfer function serves as the mathematical model used throughout the controller design and analysis.

3. Open-Loop Analysis and Motor Parameter Selection

1. Initial Motor Model

Following the derivation of the DC motor transfer function, an initial set of motor parameters was selected to construct the first mathematical model. These parameters were chosen to represent a small DC motor and to provide an initial basis for evaluating the system dynamics before proceeding with controller design.

Table 3.1: Initial Set of Motor Parameters

Parameter	Symbol	Value	Unit
Armature resistance	R	1	Ω
Armature inductance	L	0.5	H
Rotor inertia	J	0.01	Kg.m^2
Viscous friction	b	0.1	N.m.s/rad
Torque constant	K_t	0.01	N.m/A
Back EMF constant	K_e	0.01	V.s/rad

The corresponding open-loop transfer function was obtained by substituting the selected parameters into the general transfer function derived in Section 2.

2. Initial Open-Loop Response

The dynamic behavior of the initial motor model was evaluated by applying a unit-step voltage input. The resulting speed response was analyzed using MATLAB to determine its transient characteristics and steady-state gain.

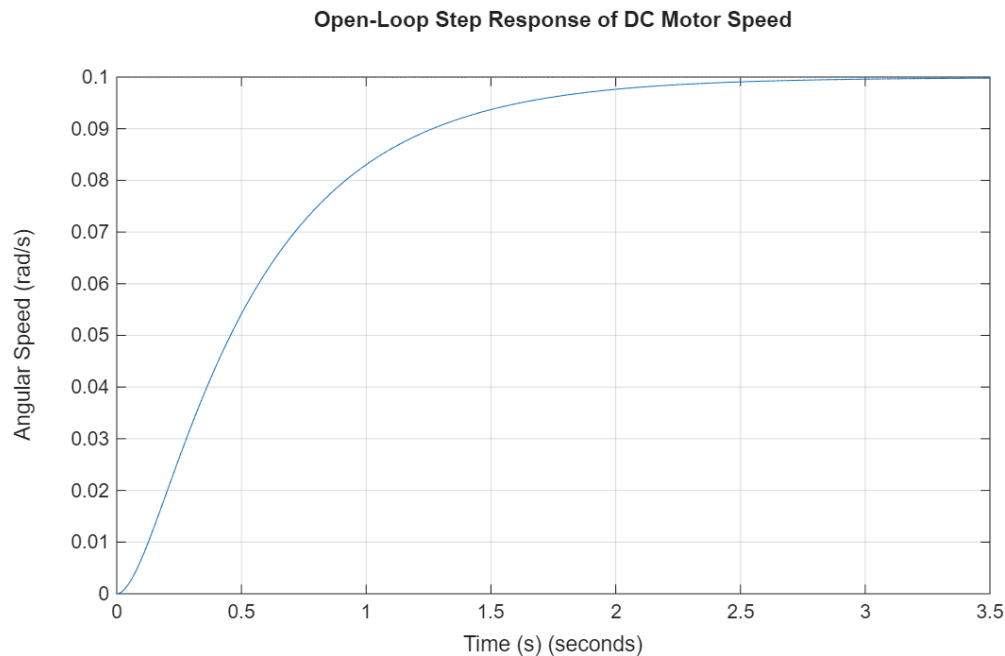


Figure 3.1: initial open-loop unit step response

Table 3.2: Initial Characteristics of the open-loop unit step response

Parameter	Value
Rise Time	1.1351 s
Settling Time	2.0652 s
Overshoot	0
Peak Value	0.0998 rad/s
Steady-State Value	0.1 rad/s

The obtained response exhibited stable second-order behavior with negligible overshoot. However, the steady-state speed achieved for a unit voltage input was considerably lower than expected for the intended light-duty speed-control application. Consequently, the resulting motor characteristics were considered unsuitable for the objectives of this project.

3. Parameter Selection

To obtain a motor model with characteristics more representative of a small light-duty DC motor, the motor parameters were revised while preserving physically meaningful values. The revised parameters resulted in a higher achievable operating speed and more realistic open-loop dynamics, providing a more suitable basis for the subsequent controller design.

The updated parameter values adopted for the remainder of this work are presented in Table 3.3.

Table 3.3: Final Set of Motor Parameters

Parameter	Symbol	Value	Unit
Armature resistance	R	1	Ω
Armature inductance	L	0.5	H
Rotor inertia	J	0.01	Kg.m ²
Viscous friction	b	0.01	N.m.s/rad
Torque constant	K_t	0.1	N.m/A
Back EMF constant	K_e	0.1	V.s/rad

These parameters were replaced in the general transfer function formula derived in section 2 to give the final plant transfer function as:

$$\frac{\Omega(s)}{V(s)} = \frac{20}{s^2 + 3s + 4}$$

4. Final Open-Loop Response

The revised motor model was analyzed under the same unit-step input to verify that its dynamic characteristics were suitable for controller design.

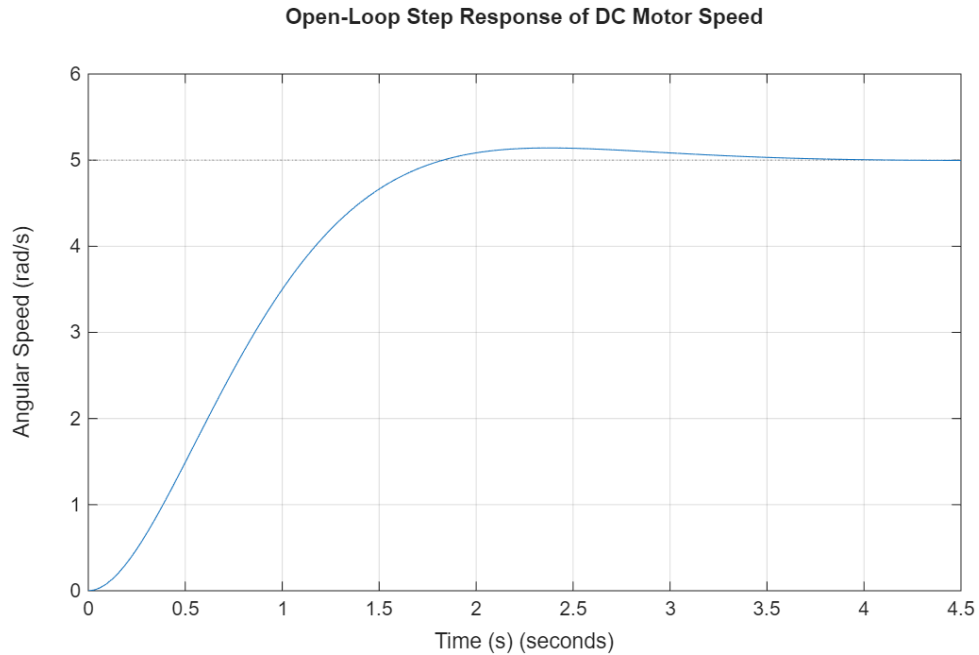


Figure 3.2: final open-loop unit step response

Table 3.4: Final Characteristics of the open-loop unit step response

Parameter	Value
Rise Time	1.1442 s
Settling Time	2.8713 s
Overshoot	2.8369 %
Peak Value	5.1418 rad/s
Steady-State Value	5 rad/s

Compared with the initial model, the revised motor exhibited a significantly higher steady-state speed while maintaining stable second-order behavior and a well-damped transient response. These characteristics better represent the intended application and provide an appropriate foundation for controller design.

5. Pole Analysis

The poles of the revised open-loop transfer function were computed to verify system stability.

The open-loop poles were found to be:

$$s_{1,2} = -1.5 \pm 1.3229j$$

Since both poles lie in the left-half of the complex plane, the open-loop system is stable. Furthermore, the complex-conjugate nature of the poles explains the underdamped response observed in the step-response simulation.

4. Controller Design

1. Design Requirements

The primary objective of the controller was to regulate the motor speed while satisfying predefined transient and steady-state performance specifications. Based on the intended application of a small light-duty DC motor, the controller was required to achieve a settling time not exceeding approximately 2 s, a maximum overshoot below 5%, zero steady-state error for a step speed reference, and stable closed-loop operation.

Although the desired settling time of the final closed-loop system was specified as 2 s, the controller was intentionally designed using a target settling time of 1.5 s. This decision was motivated by the fact that the relationships between settling time, overshoot, and dominant pole location are derived for second-order systems. Since introducing integral action increases the order of the closed-loop system, these relationships provide only an approximation of the actual response. Designing for a faster second-order equivalent compensates for the influence of the additional pole, allowing the final higher-order system to settle close to the desired 2-second specification.

These requirements led to the following values of damping ratio, natural frequency, and desired closed-loop poles:

$$\zeta = 0.7; \omega_n = 3.81 \text{ rad/s}; P_{des} = -\zeta\omega_n \pm \omega_n\sqrt{1 - \zeta^2}j = -2.7 \pm 2.7j$$

2. PI Controller Design

Because zero steady-state error was required for a step speed reference, a proportional-integral (PI) controller was first considered. The integral action increases the system type by one, eliminating the steady-state error, while the proportional gain primarily influences the transient response.

The PI controller was expressed as:

$$C_{PI}(s) = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s}$$

Initially, an analytical design approach based on coefficient matching was investigated. The desired dominant poles were determined from the transient response specifications, and the closed-loop characteristic equation was equated to the desired characteristic polynomial. However, because the introduction of the PI controller results in a third-order closed-loop system, the available equations were insufficient to uniquely determine both controller gains while simultaneously satisfying the required transient specifications. Consequently, this approach was abandoned in favor of a graphical root-locus design.

The root locus of the compensated system was then constructed, and several controller gains were investigated to evaluate the achievable closed-loop pole locations.

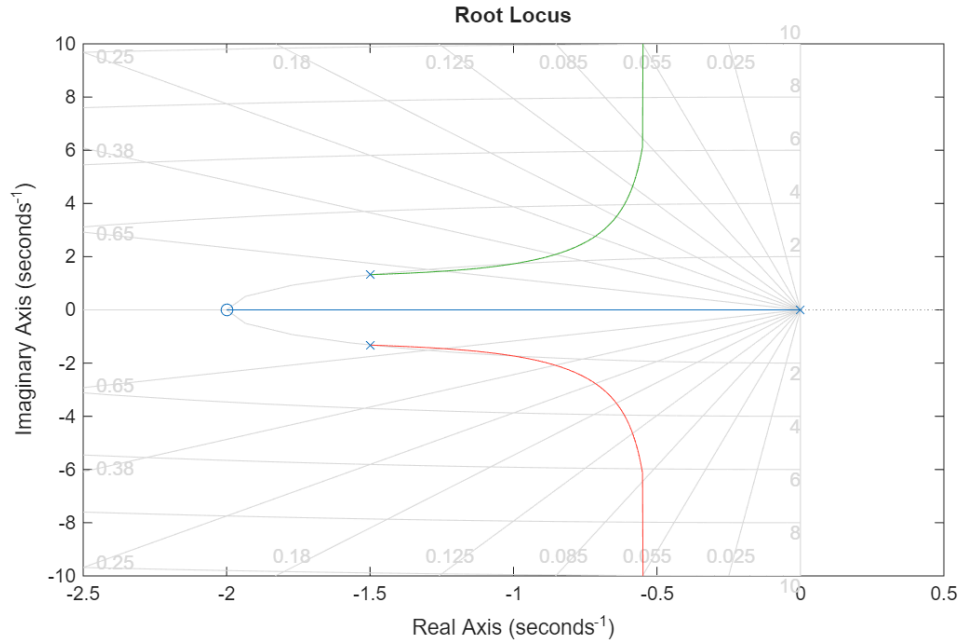


Figure 4.1: Root Locus Plot

Despite successfully eliminating the steady-state error, the PI controller alone was unable to simultaneously satisfy the desired settling time and overshoot requirements. This limitation arises because the controller provides only one adjustable zero, offering limited freedom to shape the transient response while preserving the integral action.

The resulting unit-step response is shown below.

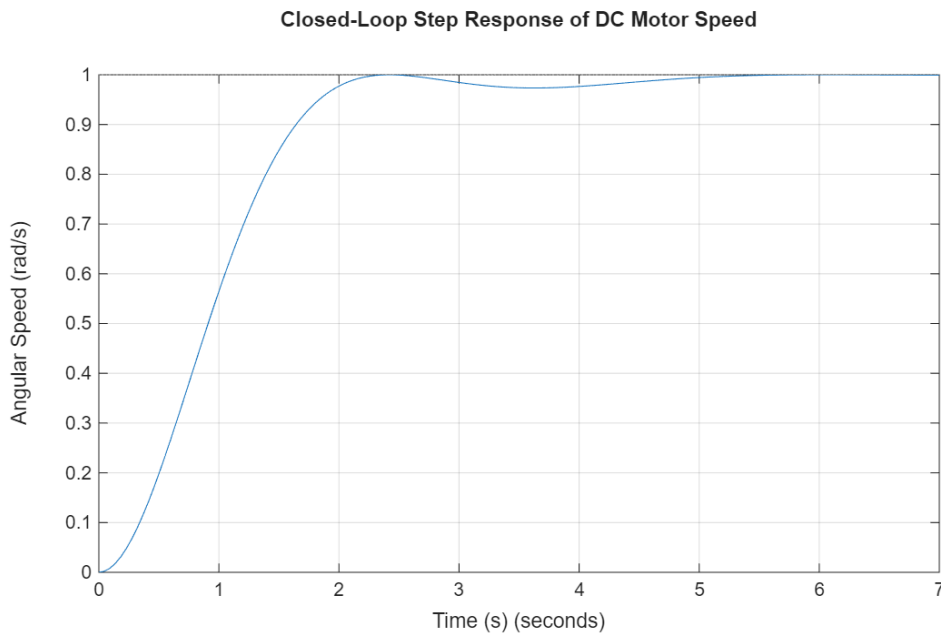


Figure 4.2: Closed-loop step response using the PI controller

The obtained response confirmed that the PI controller provided satisfactory steady-state accuracy but insufficient transient performance. Therefore, an additional compensating element was required.

3. PI + Lead Controller Design

To improve the transient response while preserving the zero steady-state error provided by the integral action, a Lead compensator was introduced in series with the PI controller. The resulting controller has the form:

$$C(s) = K \frac{(s + z_1)(s + z_2)}{s(s + p)}$$

The Lead compensator introduces an additional zero and pole, increasing the designer's flexibility in shaping the root locus and relocating the dominant closed-loop poles.

The desired dominant pole location was first determined from the target settling time of 1.5 s and the specified overshoot. The angle criterion of the root locus method was then applied to evaluate whether the desired pole belonged to the existing root locus. Since the angle condition was not satisfied, an angle of deficiency was obtained, indicating the additional phase contribution required from the Lead compensator.

The Lead pole and zero were subsequently selected so that their angular contribution supplied the required phase deficiency. Finally, the controller gain was determined using the magnitude condition of the root locus.

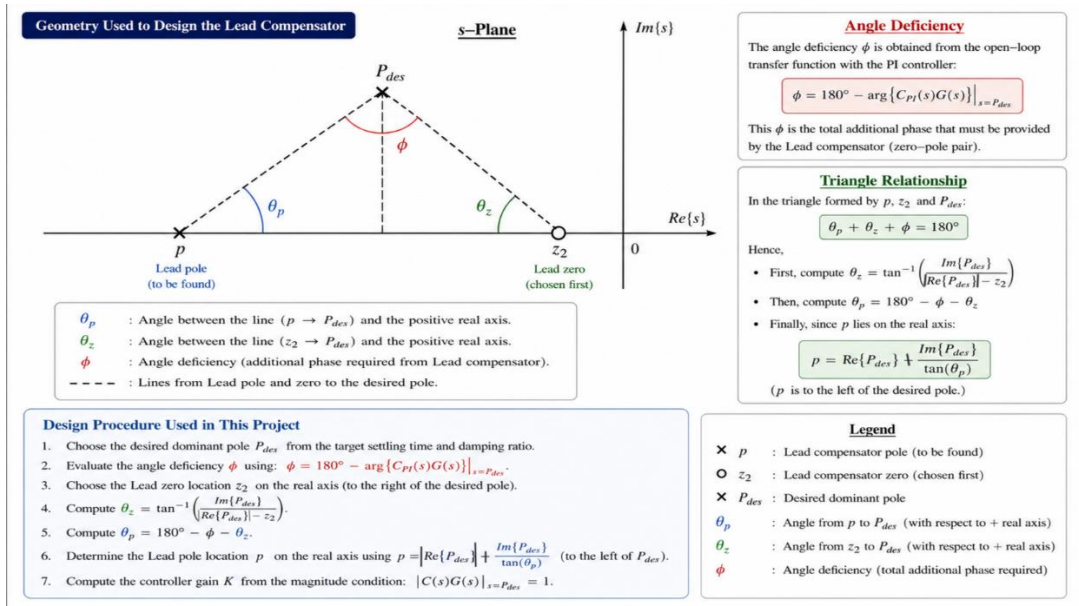


Figure 4.3: Geometry used to design the Lead compensator

The resulting PI + Lead controller was then implemented and evaluated through closed-loop simulation.

4. Controller Refinement

The initial PI + Lead controller produced a stable response with a settling time of approximately 2.18 s and an overshoot of only 1.9%. Although this response was well damped, the settling time remained slightly above the desired specification.

Rather than accepting this result, the controller parameters were further refined in order to obtain a faster response while maintaining an acceptable overshoot. After adjustment, the final controller achieved a settling time of approximately 1.95 s and a maximum overshoot of 4.29%, both of which satisfied the original design objectives.

The performance comparison between the two PI + Lead designs is summarized in Table 4.1.

Table 4.1: Comparison between the first and final PI + Lead controller designs

Design	Settling Time (s)	Overshoot (%)	Selected
First PI + Lead	2.1817	1.8966	NO
Final PI + Lead	1.9526	4.2891	YES

The second design was therefore adopted for the remainder of this work, as it achieved a better compromise between response speed and transient overshoot. The final controller adopted had the following form:

$$C(s) = 4.9307 \frac{(s + 2.5)(s + 2)}{s(s + 27.99)}$$

5. Closed-Loop Verification

The final PI + Lead controller was verified by analyzing the closed-loop pole locations and the corresponding unit-step response. All closed-loop poles were located in the left-half of the complex plane, confirming the stability of the controlled system. The designed complex-conjugate pole pair was successfully placed at the desired location, satisfying the root-locus design objective. However, an additional real pole remained closer to the imaginary axis than anticipated, causing the actual settling time to differ slightly from the value predicted by the second-order design equations. This observation is consistent with the fact that the closed-loop system is fourth-order, whereas the classical relationships between settling time, overshoot, and pole location are based on second-order system approximations. Consequently, the final controller performance was validated through MATLAB simulations rather than relying solely on theoretical estimates.

Closed-Loop poles: -24.1921; -2.7000 + 2.7000j; -2.7000 - 2.7000j; -1.3979.

The closed-loop step response is presented below.

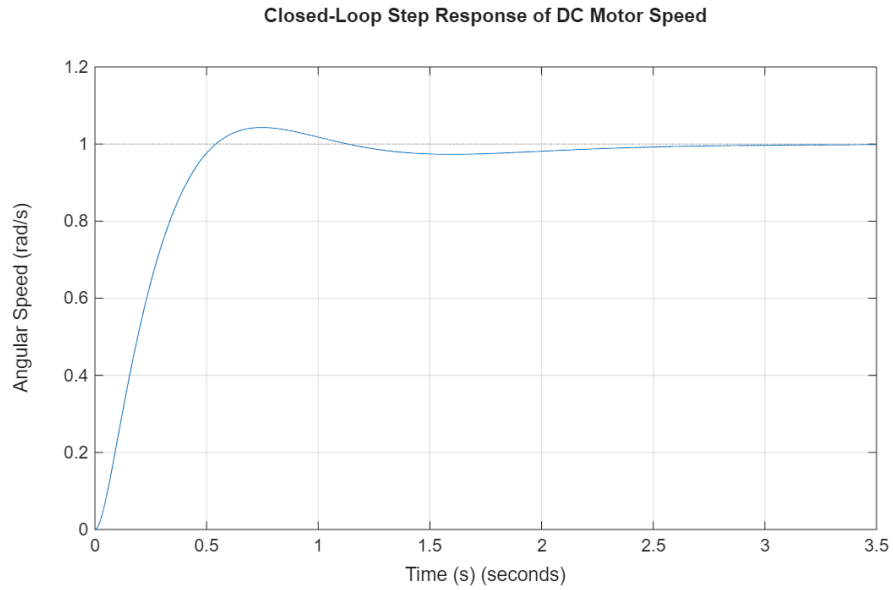


Figure 4.4: Final closed-loop unit-step response using the PI + Lead controller

The corresponding transient characteristics obtained from MATLAB confirmed that the controller satisfied the intended design objectives, providing a settling time close to 2 s, an overshoot below 5%, and zero steady-state error.

In addition to the output response, the controller output (control effort) was examined to verify that the required armature voltage remained physically reasonable for the assumed motor supply (± 12 Volts range).

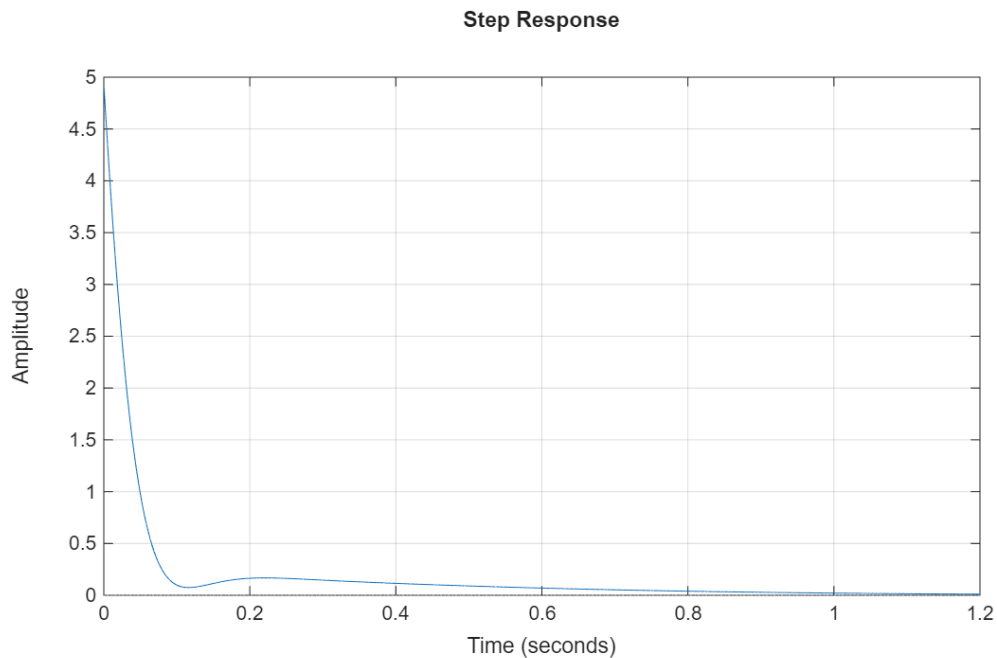


Figure 4.5: Controller output (control effort) for the final PI + Lead controller

The controller output remained within acceptable operating limits, indicating that the designed controller is suitable for the assumed motor characteristics and operating conditions under nominal linear operation.

6. Tracking of a Time-Varying Reference

Although the controller was designed based on a step speed command, many practical DC motor applications require the reference speed to vary continuously with time rather than remain constant. Examples include speed scheduling in conveyor systems, fan speed regulation, and robotic actuators following predefined motion profiles. Therefore, an additional simulation was performed to evaluate the tracking capability of the designed controller for a continuously varying reference signal.

Following the initial regulation of the motor speed to a constant reference of 40 rad/s, the reference input was changed at $t = 10\text{s}$ to the sinusoidal signal:

$$\omega_{ref}(t) = 40 + 5 \sin(0.5t)$$

which varies smoothly between 35 rad/s and 45 rad/s. The sinusoidal command was introduced only after the initial transient had completely settled, allowing the tracking performance to be evaluated independently from the startup response.

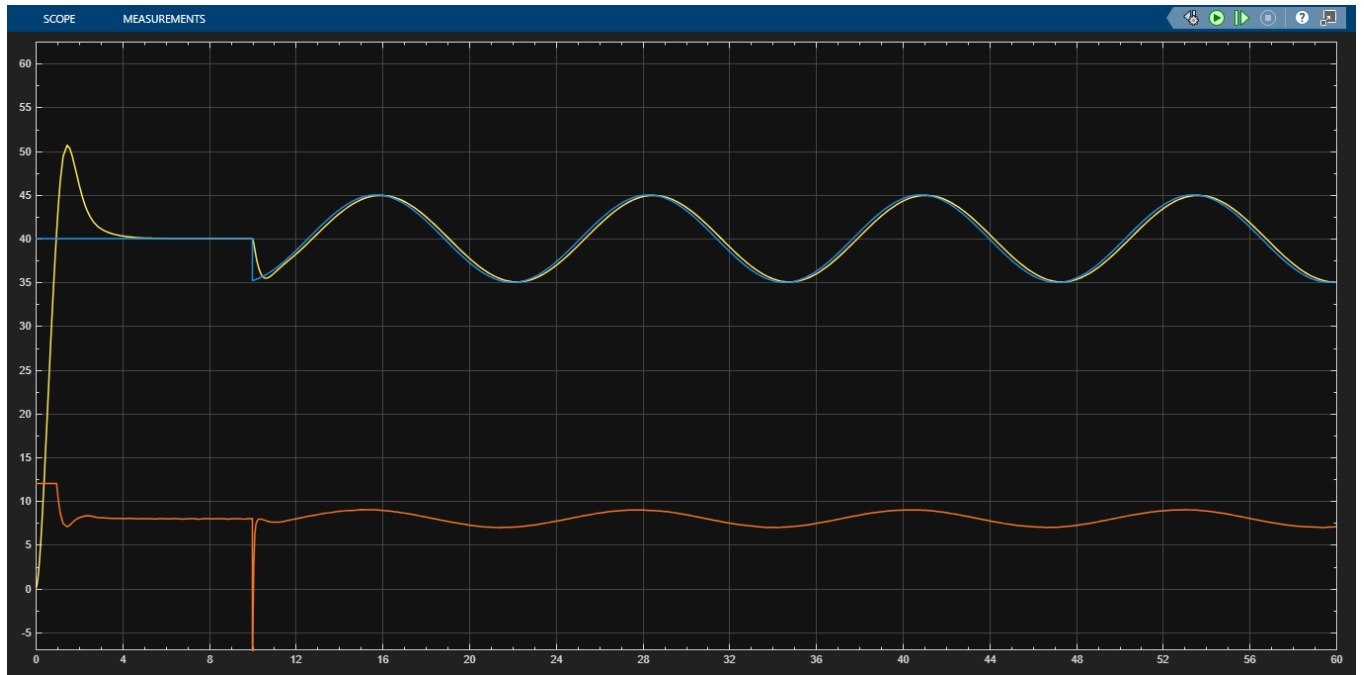


Figure 4.6: Tracking response for a sinusoidal speed reference following an initial step to 40 rad/s.

As shown in Figure 4.6, the proposed PI + Lead controller successfully tracked the sinusoidal reference throughout the simulation. After a brief transient caused by the transition from the constant to the sinusoidal reference, the motor speed closely followed

the desired command with only a small phase lag and negligible amplitude attenuation. These characteristics indicate that the controller possesses sufficient bandwidth to regulate the motor speed under slowly varying operating conditions.

The controller output varied smoothly in response to the changing speed demand and remained within the available ± 12 V supply limits during the tracking phase. This indicates that the desired speed profile can be achieved without requiring excessive control effort or actuator saturation.

Overall, this additional simulation demonstrates that the designed controller is not limited to step-reference regulation but is also capable of accurately tracking realistic time-varying speed commands. This result further validates the suitability of the proposed controller for practical speed-control applications where the desired motor speed changes continuously over time.

5. Disturbance Rejection Analysis

1. Load Torque Disturbance Model

In practical applications, DC motors rarely operate under constant mechanical loading. Variations in external loads, such as changes in payload, friction, or mechanical resistance, produce disturbance torques that tend to reduce the motor speed. Therefore, evaluating the controller under load disturbances provides a more realistic assessment of its performance than reference tracking alone.

To investigate disturbance rejection, the mathematical model of the motor was extended to include an external load torque input $T_L(t)$. The mechanical equation becomes:

$$J\dot{\omega}(t) + b\omega(t) = K_t i(t) - T_L(t)$$

where the disturbance torque opposes the electromagnetic torque generated by the motor.

Following the same derivation procedure presented in Section 2, the transfer function relating the disturbance torque to the motor angular speed is obtained as:

$$G_d(s) = \frac{\Omega(s)}{T_L(s)} = \frac{-(Ls + R)}{LJs^2 + (RJ + Lb)s + (Rb + K_t K_e)}$$

The negative sign indicates that an increase in load torque reduces the motor speed.

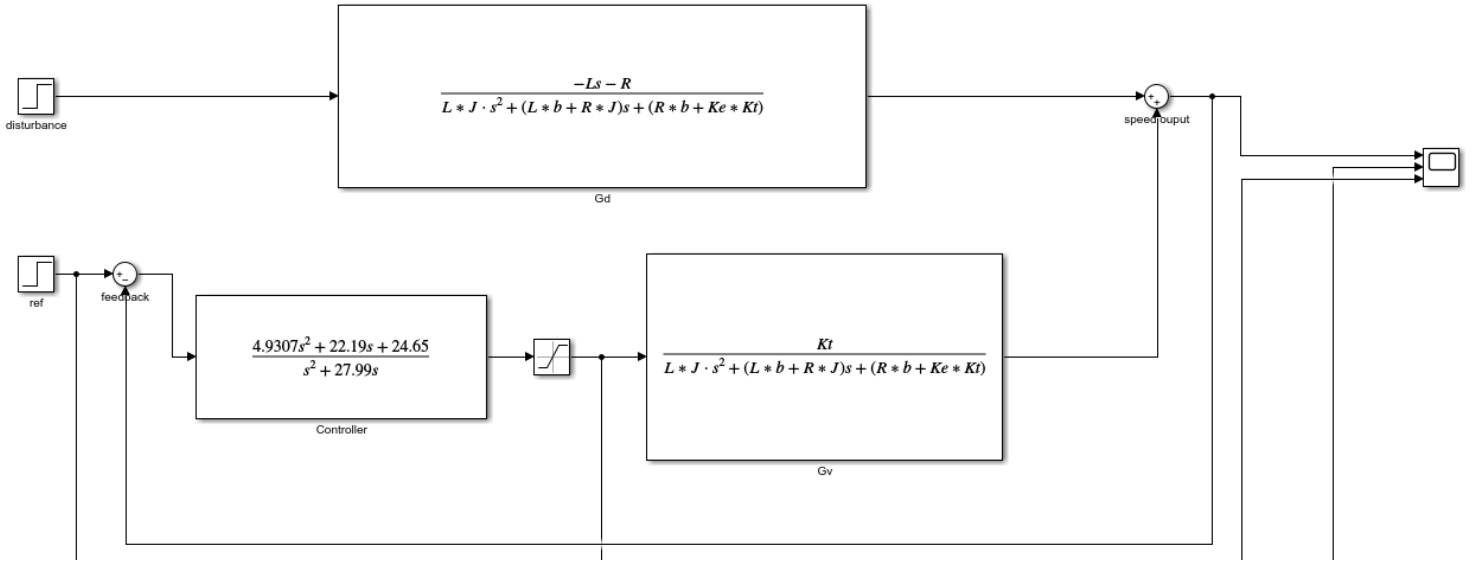


Figure 5.1: Complete Simulink model used for disturbance rejection analysis.

2. Simulation Conditions

The disturbance rejection capability of the proposed controller was evaluated through time-domain simulations in Simulink.

The following operating conditions were adopted:

- Reference speed: 50 rad/s
- Load disturbance: step disturbance applied at $t = 10$ s.
- Controller output: saturated at: ± 12 V to represent the assumed motor supply voltage.

Three disturbance magnitudes were investigated in order to evaluate the controller under progressively increasing mechanical loading.

3. Response to a 0.1 N·m Load Disturbance

The first simulation considered a moderate disturbance torque of $T_L = 0.1 \text{ N.m}$. The disturbance caused a temporary reduction in motor speed immediately after its application. However, the controller rapidly increased the armature voltage to compensate for the additional mechanical load, allowing the motor speed to return to its reference value without steady-state error. The transient speed deviation remained relatively small, and the controller output stayed within the allowable voltage limits throughout the simulation. These results demonstrate that the proposed controller provides satisfactory disturbance rejection for disturbances representative of normal operating conditions.

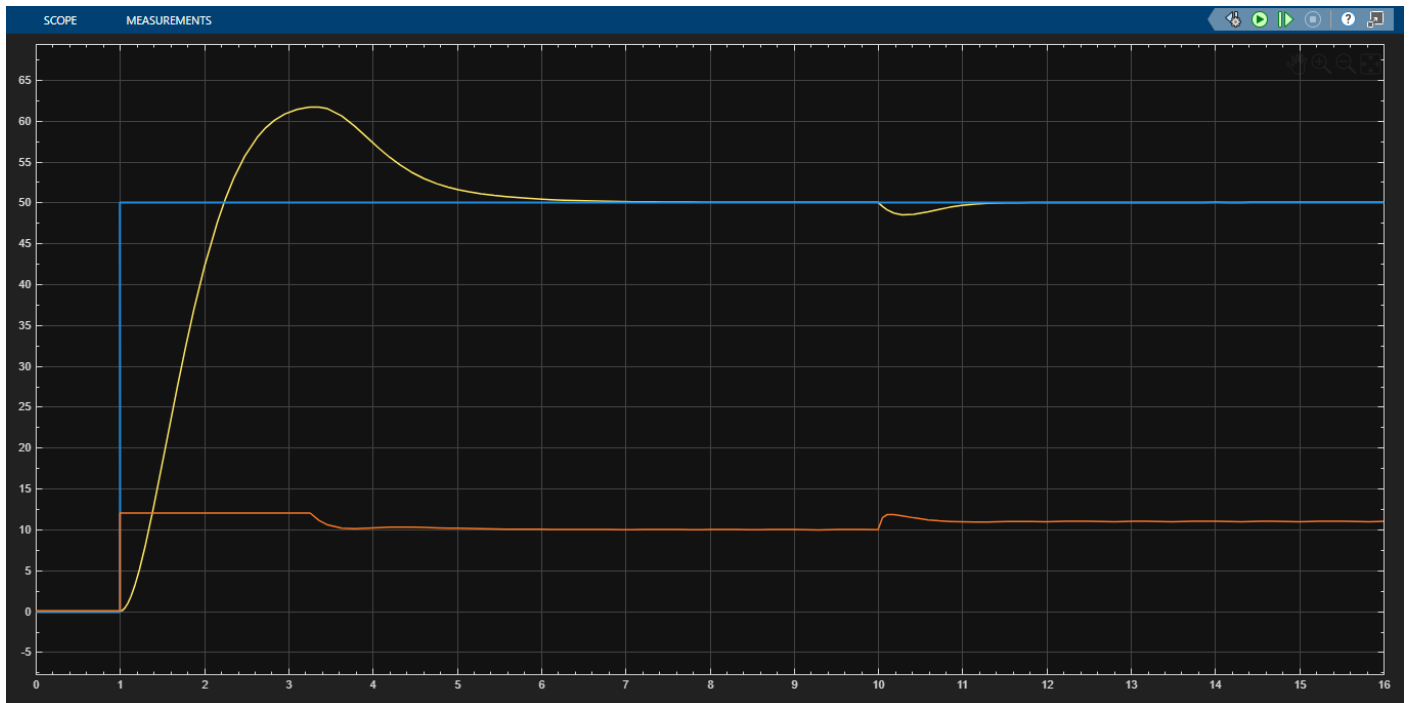


Figure 5.2: Response to a 0.1 N·m load disturbance (50 rad/s reference, ± 12 V controller limit).

4. Response to a 0.2 N·m Load Disturbance

The disturbance magnitude was then increased to **0.2 N·m**. As expected, the larger disturbance produced a greater temporary reduction in motor speed and required a higher control effort. Nevertheless, the controller remained capable of restoring the motor speed to the desired reference without steady-state error.

Compared with the previous case, the recovery process became slower due to the increased mechanical load. However, the closed-loop system remained stable and maintained satisfactory regulation, indicating that the designed controller retains acceptable performance under moderately increased loading conditions.

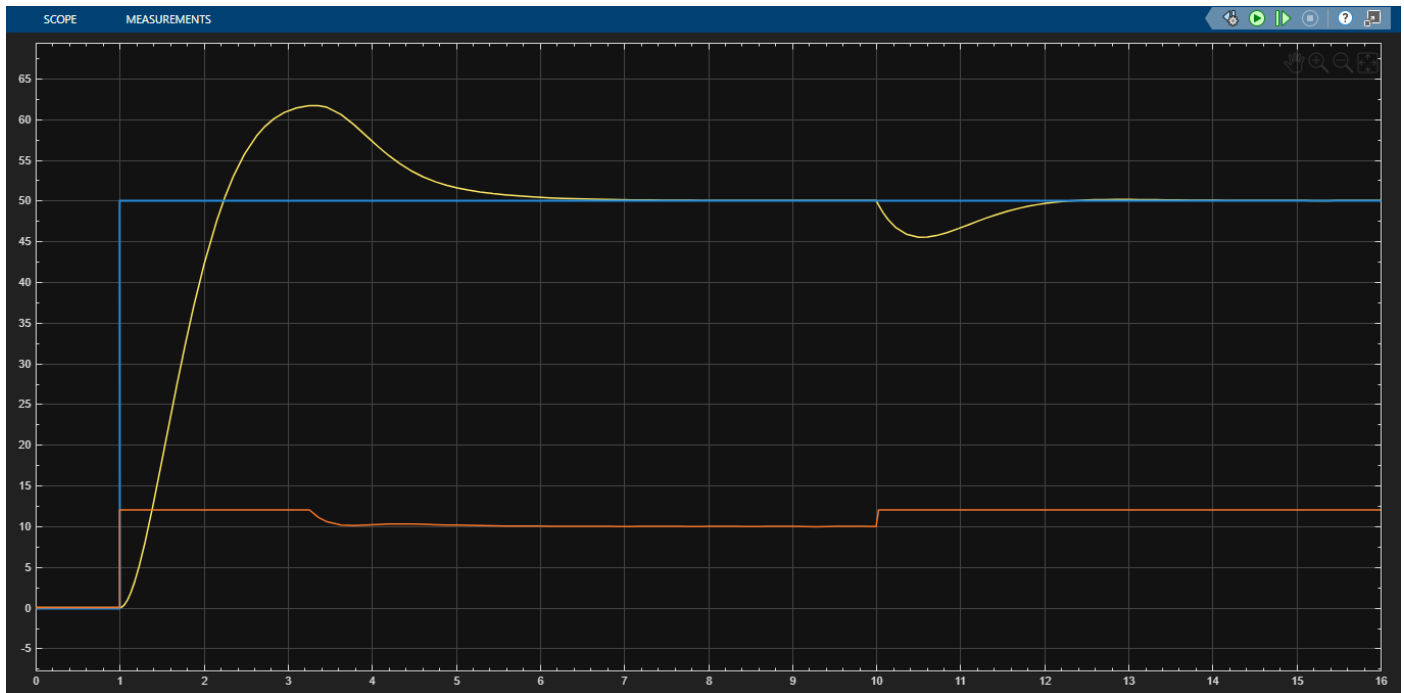


Figure 5.3: Response to a $0.2 \text{ N}\cdot\text{m}$ load disturbance (50 rad/s reference, $\pm 12 \text{ V}$ controller limit).

5. Response to a $0.4 \text{ N}\cdot\text{m}$ Load Disturbance

Finally, a severe disturbance torque of $0.4 \text{ N}\cdot\text{m}$ was applied to evaluate the controller under extreme loading conditions.

Unlike the previous cases, the controller was unable to completely recover the motor speed to its reference value. Although the controller attempted to compensate by increasing the armature voltage, the voltage quickly reached the imposed saturation limit of $\pm 12 \text{ V}$. Since no additional voltage could be supplied, the motor was incapable of generating sufficient torque to overcome the applied load.

Consequently, the motor speed settled below the desired reference despite the presence of integral action in the controller.

This behavior should not be interpreted as a failure of the controller design. Rather, it reflects the physical limitation imposed by the available supply voltage and the torque capability of the selected motor. Once the actuator reaches saturation, no linear controller can provide additional control authority without increasing the available supply voltage or selecting a motor with greater torque capability.

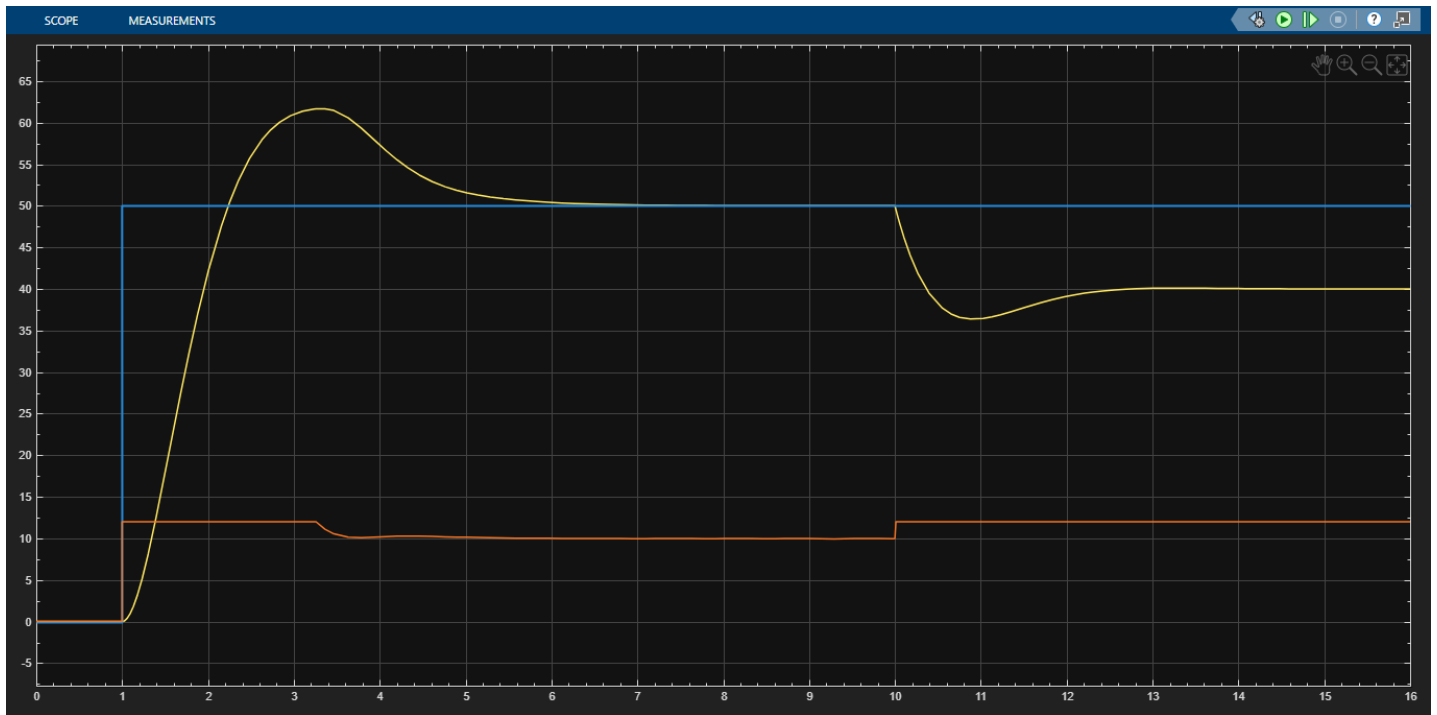


Figure 5.4: Response to a 0.4 N·m load disturbance (50 rad/s reference, ± 12 V controller limit).

6. Discussion

The disturbance rejection study demonstrates that the designed PI + Lead controller successfully maintains speed regulation under realistic operating disturbances while preserving closed-loop stability and zero steady-state error whenever sufficient actuator authority is available.

For disturbance torques of 0.1 N·m and 0.2 N·m, the controller effectively compensated for the additional load by increasing the applied armature voltage, restoring the motor speed to its reference after a short transient period. As the disturbance magnitude increased, the transient speed deviation became larger and the recovery time increased, which is consistent with the expected behavior of a feedback control system.

For the 0.4 N·m disturbance, the controller reached the actuator saturation limit before sufficient compensating torque could be produced. The resulting steady-state speed error therefore originated from the physical limitations of the motor-drive system rather than from instability or inadequate controller design.

These results indicate that the proposed controller provides satisfactory disturbance rejection within the practical operating range of the selected motor while also highlighting the importance of considering actuator limitations during controller validation.

Additional simulations performed with an increased actuator limit of ± 24 V resulted in a significantly improved recovery, confirming that the degraded performance observed

under the 0.4 N·m disturbance was primarily caused by actuator saturation rather than deficiencies in the controller itself.

6. Robustness Analysis

1. Objective of the Robustness Test

After verifying the nominal closed-loop performance and disturbance rejection capability, the robustness of the designed PI + Lead controller was evaluated. In practical systems, motor parameters are rarely known exactly. Parameters such as inertia, friction, armature resistance, and torque constant may vary due to load changes, heating, manufacturing tolerances, or operating conditions. Therefore, a controller should not only perform well for the nominal model, but should also maintain acceptable performance when the plant parameters deviate from their assumed values.

In this section, the controller designed in Section 4 was kept unchanged, while selected motor parameters were varied individually. The purpose of this test was to determine whether the controller could maintain stability, zero steady-state error, and acceptable transient response under moderate parameter variations.

The nominal motor parameters were modified according to the following cases: $J = 1.2 \cdot J$; $b = 1.5 \cdot b$; $K_t = 0.8 \cdot K_t$; $R = 1.2 \cdot R$.

These variations were selected to represent realistic changes in motor operation. Increasing J represents a heavier rotating load, increasing b represents higher mechanical friction, decreasing K_t represents a weaker torque-producing motor, and increasing R represents the effect of armature heating or increased winding resistance.

2. Open-Loop Response Under Parameter Variations

Before evaluating the controlled system, the open-loop response was examined for each parameter variation. This provided a baseline for understanding how the motor dynamics were affected before the controller was applied.

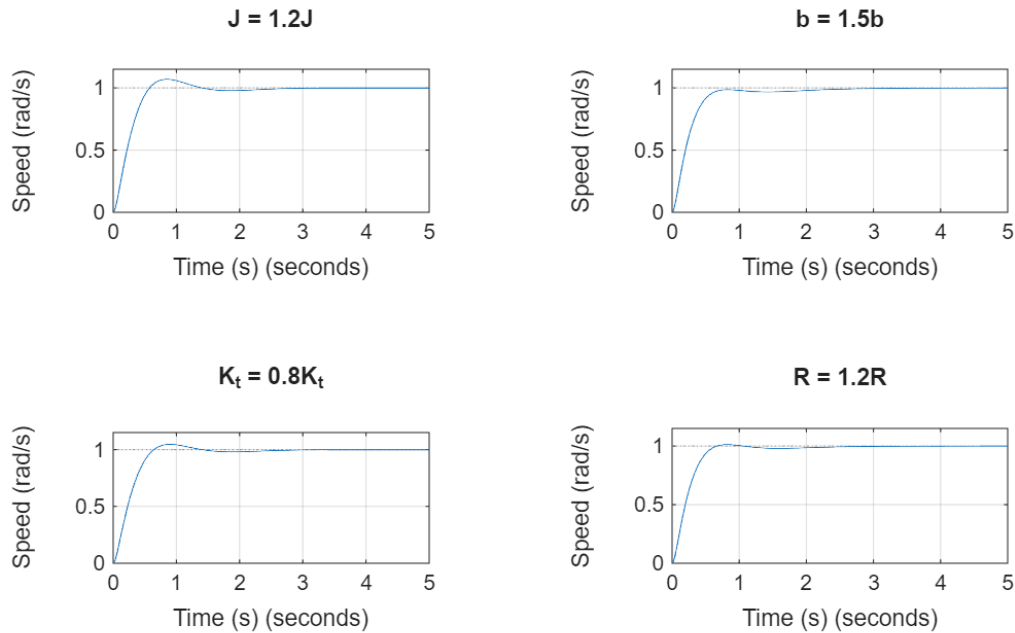
Table 6.1: Open-loop performance under parameter variations

Parameter Variation	Rise Time (s)	Settling Time (s)	Overshoot (%)	Open-Loop Poles
J = 1.2*J	1.3032	2.9095	2.0975	$-1.4167 \pm 1.1517j$
b = 1.5*b	1.0749	1.6208	1.9258	$-1.7500 \pm 1.3919j$
K_t = 0.8*K_t	1.2822	1.9413	1.7322	$-1.5000 \pm 1.1619j$
R = 1.2*R	1.1955	1.8308	1.2953	$-1.7000 \pm 1.2288j$

The open-loop results show that all modified systems remained stable, since all poles were located in the left-half of the complex plane. However, the transient response and steady-state speed changed depending on the modified parameter. The increase in inertia produced the slowest open-loop response, as expected, because a larger inertia resists changes in speed. Increasing the viscous friction moved the open-loop poles farther to the left and increased damping, resulting in a faster settling response but a lower steady-state speed. Variations in K_t and R also affected the motor gain and dynamics, confirming that the uncontrolled motor performance is sensitive to parameter changes.

3. Closed-Loop Response Under Parameter Variations

The same parameter variations were then applied to the closed-loop system while keeping the PI + Lead controller unchanged. This test evaluates whether the controller remains effective when the plant differs from the nominal design model.



The resulting closed-loop performance metrics are summarized in Table 6.2.

Table 6.2: Closed-loop performance under parameter variations

Figure 6.1: Closed-loop step responses under parameter variations

Parameter Variation	Rise Time (s)	Settling Time (s)	Overshoot (%)	Open-Loop Poles	Evaluation
Nominal	—	1.9526	4.2891	-24.1921, -2.7000±2.7000j, -1.3979	Meets requirements
J = 1.2*J	0.3937	2.0367	6.9302	-24.9310, -2.2320±2.5605j, -1.4284	Stable, but overshoot exceeds requirement
b = 1.5*b	0.4196	2.0262	0	-24.0938, -3.0273±2.4678j, -1.3415	Stable, well damped
K_t = 0.8*K_t	0.4290	1.9186	4.6307	-25.0450, -2.2680±2.4565j, -1.4090	Stable and acceptable
R = 1.2*R	0.3950	1.6771	1.1885	-24.1104, -2.9248±2.3970j, -1.4301	Stable and acceptable

All tested closed-loop systems remained stable, with all poles located in the left-half of the complex plane. The controller also maintained zero steady-state error in all cases due to the integral action included in the PI + Lead structure. This confirms that the controller preserves the main steady-state objective even when the motor parameters vary.

The most critical case was the increase in rotor inertia. When J was increased by 20%, the settling time increased slightly to approximately 2.04 s, while the overshoot increased to approximately 6.93%. This exceeds the original overshoot requirement of 5%, indicating that the controller is somewhat sensitive to increases in inertia. This result is physically reasonable because a larger inertia makes the motor harder to accelerate and decelerate, reducing the effectiveness of the nominal controller tuning.

When the viscous friction coefficient was increased by 50%, the response became more damped and exhibited no overshoot. The settling time was approximately 2.03 s, which is only slightly above the 2-second target. This case remained practically acceptable because the response was stable, smooth, and well damped.

When the torque constant was reduced by 20%, the motor became less effective at producing torque for a given armature current. Nevertheless, the controller maintained acceptable performance, with a settling time of approximately 1.92 s and an overshoot of approximately 4.63%. This result indicates that the controller can tolerate a moderate reduction in motor torque constant while still satisfying the main design requirements.

The increase in armature resistance produced the best closed-loop response among the tested variations, with a settling time of approximately 1.68 s and an overshoot of only 1.19%. Although increased resistance generally reduces the current available for a given voltage, the closed-loop controller was able to compensate effectively under the tested conditions.

4. Control Effort Under Parameter Variations

The controller output was also examined for each parameter variation. The control effort plots were found to be similar across the tested cases, and no major abnormal behavior was observed. Therefore, separate controller output figures are not included in this section to avoid unnecessary repetition.

Instead, the control effort analysis is summarized qualitatively. The controller remained within reasonable operating limits for the tested unit-step responses, indicating that the robustness results were achieved without requiring unrealistic control action.

5. Robustness Discussion

The robustness analysis shows that the PI + Lead controller maintains stable closed-loop operation under all tested parameter variations. The system also preserves zero steady-

state error in all cases, confirming the importance of the integral component in the controller structure.

However, the results also show that the controller is not equally insensitive to all parameter changes. The response was most affected by an increase in rotor inertia, where the overshoot exceeded the nominal design requirement. This suggests that the controller has limited robustness margin with respect to inertia variations. In practical applications, this would be important if the motor were expected to drive loads with significantly changing inertia.

On the other hand, the controller handled variations in friction, torque constant, and armature resistance reasonably well. The responses remained stable and close to the desired performance range, with only small deviations from the nominal specifications.

Overall, the controller can be considered **moderately robust**. It performs well for moderate variations in most motor parameters, but its performance is more sensitive to changes in inertia. This limitation does not invalidate the controller design; rather, it identifies the range of operating conditions for which the controller is most reliable and highlights a possible direction for future improvement.

6. Conclusion of Robustness Analysis

The robustness study confirmed that the designed PI + Lead controller remains stable and maintains zero steady-state error under all tested parameter variations. The controller provided acceptable performance for changes in friction, torque constant, and armature resistance. However, a 20% increase in rotor inertia caused the overshoot to exceed the original design requirement, indicating that inertia variation is the most critical uncertainty for this controller.

These results demonstrate that the controller is suitable for the intended light-duty motor application, provided that large variations in load inertia are not expected. For applications involving significant inertia changes, further controller refinement or adaptive control techniques may be required.

7. Discussion and Limitations

The results obtained throughout this project demonstrate that the proposed PI + Lead controller successfully achieved the primary objective of regulating the speed of the DC motor while satisfying the predefined design requirements under nominal operating conditions. Starting from a mathematical model derived from the electrical and mechanical governing equations of the motor, a systematic controller design procedure was followed using classical control techniques. The final controller provided stable closed-loop operation, zero steady-state error for step speed commands, and transient characteristics that closely matched the desired specifications.

The controller design also illustrates an important aspect of classical control engineering. Although the dominant closed-loop poles were selected using second-order design relationships, the addition of the PI controller increased the order of the closed-loop system. Consequently, the second-order settling time and overshoot formulas served only as an initial approximation. This explains why the controller was intentionally designed using a target settling time of 1.5 s instead of the required 2 s, allowing the final fourth-order system to satisfy the desired transient performance after simulation-based verification.

The disturbance rejection analysis further demonstrated the effectiveness of the controller under practical operating conditions. For moderate load disturbances, the controller successfully compensated for the additional mechanical load by increasing the applied armature voltage and restoring the motor speed to its reference value without steady-state error. As the disturbance magnitude increased, the transient speed deviation became larger and the recovery time increased, which is consistent with the expected behavior of a feedback control system.

The analysis of large disturbance torques highlighted an important practical consideration. Under a disturbance torque of $0.4 \text{ N}\cdot\text{m}$, the controller was unable to restore the motor speed to its reference because the required control voltage exceeded the available $\pm 12 \text{ V}$ supply limit. The resulting performance degradation was therefore caused by actuator saturation rather than deficiencies in the controller itself. Additional simulations performed with an increased voltage limit confirmed that the controller recovered the reference speed more effectively once sufficient control authority became available. This observation emphasizes that the achievable disturbance rejection capability depends not only on the controller design but also on the physical capabilities of the actuator and motor.

The robustness analysis showed that the controller maintained closed-loop stability and zero steady-state error for all investigated parameter variations. Changes in viscous friction, armature resistance, and torque constant produced only moderate changes in the transient response, indicating that the controller possesses satisfactory robustness for the intended application. However, the increase in rotor inertia resulted in an overshoot slightly exceeding the original design requirement, demonstrating that inertia variation is the most influential parameter among those considered in this study. Although the system remained stable, this result indicates that the controller tuning is somewhat sensitive to significant increases in mechanical inertia.

Despite the encouraging simulation results, several limitations should be recognized. The controller was designed using a linear mathematical model that neglects several nonlinear effects commonly encountered in practical DC motors, including Coulomb friction, magnetic saturation, backlash, and parameter variations caused by temperature.

Furthermore, the study was entirely simulation-based and did not include experimental validation on real hardware. Additional practical effects such as sensor noise, encoder quantization, PWM switching dynamics, and communication delays were also neglected. Consequently, the obtained results represent the expected performance of an idealized system rather than a complete representation of a physical implementation.

Nevertheless, these limitations do not diminish the educational and engineering value of the project. The work demonstrates a complete classical control design workflow, beginning with mathematical modeling and continuing through controller design, closed-loop verification, disturbance rejection analysis, actuator saturation assessment, and robustness evaluation. This systematic approach provides a solid foundation for future hardware implementation and more advanced controller development.

8. Conclusion and Future Work

This project presented the complete design and evaluation of a classical PI + Lead controller for DC motor speed regulation using MATLAB and Simulink. Beginning with a mathematical model derived from the motor's electrical and mechanical governing equations, a systematic controller design procedure was followed, including open-loop analysis, controller synthesis, closed-loop verification, disturbance rejection evaluation, and robustness analysis.

The final controller successfully achieved the primary design objectives by providing stable closed-loop operation, zero steady-state error for step speed commands, and transient performance close to the desired specifications. The controller also demonstrated satisfactory disturbance rejection for moderate load torques and maintained acceptable performance under several motor parameter variations, confirming its suitability for the intended light-duty speed-control application.

The project also highlighted the importance of practical engineering considerations beyond theoretical controller design. Actuator voltage saturation was shown to significantly influence the achievable disturbance rejection performance, while robustness analysis identified rotor inertia as the parameter having the greatest impact on the closed-loop response. These observations emphasize that successful controller design requires consideration of both mathematical models and physical system limitations.

Future work may include implementing the controller on a real DC motor using a microcontroller, incorporating sensor feedback and PWM motor drive dynamics, and introducing anti-windup techniques to improve performance under actuator saturation. More advanced control strategies, such as state-space control, optimal control, or

adaptive control, may also be investigated to further improve robustness and performance under varying operating conditions.